

Equations

dT/dt = growth of tumor cells - tumor cells inhibited by NK cells (ADCC-mediated NK activation is now moved into dNK/dt) - tumor cells inhibited by CTL cells - tumor cells inhibited by effector B cells - death of tumor cells

dM/dt = circulating MDSC level + MDSC recruitment in presence of tumor - MDSC death

dNK/dt = circulating NK level + NK recruitment in presence of tumor + NK functional activation supported by effector B cells via ADCC - NK inhibited by MDSC - NK death

$dCTL/dt$ = CTL stimulation by tumor-NK interaction + CTL recruitment in presence of tumor + CTL support via effector B-cell APC/costimulation - CTL inhibited by MDSC - CTL death

dB/dt = circulating B level + B recruitment in presence of tumor - B inhibited by MDSC - B death

Original CIR equation

Original CIR ODE core ($\tau = 0$):

$$\frac{dx_T}{dt} = \alpha_1 x_T \ln\left(\frac{\eta}{x_T}\right) - \beta_1 x_T x_{NK} - \beta_2 x_T x_{CTL} - \zeta_1 x_T$$

$$\frac{dx_M}{dt} = \alpha_2 + \alpha_3 \frac{x_T}{\gamma_1 + x_T} - \zeta_2 x_M$$

$$\frac{dx_{NK}}{dt} = \alpha_4 + \alpha_5 \frac{x_T^2}{\gamma_2 + x_T^2} - \beta_3 x_M x_{NK} - \zeta_3 x_{NK}$$

$$\frac{dx_{CTL}}{dt} = \alpha_6 x_T x_{NK} + \alpha_7 \frac{x_T^2}{\gamma_3 + x_T^2} - \beta_4 x_M x_{CTL} - \zeta_4 x_{CTL}$$

CIR model ODEs with B-cell extensions

Main simulation model after feedback: B → NK activation is placed directly in the NK equation. CTL option A uses a simple direct B-only APC/costimulation term; CTL option B is retained below for comparison.

Updated ODE system (main simulation model; CTL Option A):

$$\frac{dx_T}{dt} = \alpha_1 x_T \ln\left(\frac{\eta}{x_T}\right) - \beta_1 x_T x_{NK} - \beta_2 x_T x_{CTL} - \beta_6 x_T x_B - \zeta_1 x_T$$

$$\frac{dx_M}{dt} = \alpha_2 + \alpha_3 \frac{x_T}{\gamma_1 + x_T} - \zeta_2 x_M$$

$$\frac{dx_{NK}}{dt} = \alpha_4 + \alpha_5 \frac{x_T^2}{\gamma_2 + x_T^2} \left(1 + \alpha_{10} \frac{x_B}{\gamma_5 + x_B}\right) - \beta_3 x_M x_{NK} - \zeta_3 x_{NK}$$

$$\frac{dx_{CTL}}{dt} = \alpha_6 x_T x_{NK} + \alpha_7 \frac{x_T^2}{\gamma_3 + x_T^2} + \alpha_{11} \frac{x_B}{\gamma_5 + x_B} - \beta_4 x_M x_{CTL} - \zeta_4 x_{CTL}$$

$$\frac{dx_B}{dt} = \alpha_8 + \alpha_9 \frac{x_T^2}{\gamma_4 + x_T^2} - \beta_5 x_M x_B - \zeta_5 x_B$$

$$x_B(0) = \frac{\alpha_8}{\zeta_5 + \beta_5 \alpha_2 / \zeta_2}$$

Alternate CTL option suggestion (or-gated B×T APC term):

$$\frac{dx_{CTL}}{dt} = \alpha_6 x_T x_{NK} + \alpha_7 \frac{x_T^2}{\gamma_3 + x_T^2} + \alpha_{11} \left(\frac{x_T^2}{\gamma_3 + x_T^2}\right) \left(\frac{x_B}{\gamma_5 + x_B}\right) - \beta_4 x_M x_{CTL} - \zeta_4 x_{CTL}$$

Equation-note: all non-B-cell parameters are taken directly from Kreger et al. (2023), Table 1. New B-cell parameters are listed in Table 2 below.

Table 2. New B-cell-related parameters and ranges used for sensitivity / stability scans.

Notation	Description	Value	Units	Reference	Range
$x_B(0)$	initial condition for effector B cells	tumor-free steady state $\alpha_8 / (\zeta_5 + \beta_5 \alpha_2 / \zeta_2)$	—	CIR initializes other compartments via tumor-free equilibrium logic	—
α_8	B-cell circulation / recruitment rate (modeled analogously to NK circulation, α_4).	varies	day ⁻¹	[1,2]	$[10^3, 10^5]$
α_9	Tumor-driven B-cell recruitment coefficient.	varies	day ⁻¹	[1]	$[10^{(-2)}, 10^{(-1)}]$
γ_4	Steepness parameter for tumor-driven B recruitment term.	varies	—	[1]	$[10^6, 10^8]$
β_5	Effector B-cell inhibition rate by MDSCs.	varies	day ⁻¹	[1,3]	For $\beta_5 * X_M(1)$: $[5 * 10^{(-2)}, 10^{(-1)}]$ Or $[10^{(-5)}, 5 * 10^{(-4)}]$
ζ_5	B-cell death rate.	varies (TBD)	day ⁻¹	[2]	$[10^{(-2)}, 10^{(-1)}]$
α_{10}	B-mediated enhancement of NK cytotoxicity via ADCC.	varies	—	[6-8]	[0, 8]

Notation	Description	Value	Units	Reference	Range
γ_5	B half-saturation constant for B-mediated effects ($x_B/(\gamma_5+x_B)$).	varies	—	[4,5]	$[10^2, 10^3]$
α_{11}	CTL expansion via B as APC/costimulation.	varies	day ⁻¹	[1,9,10]	$[5 \cdot 10^{(-2)}, 1]$
β_6	tumor inhibition rate by effector B.	varies	day ⁻¹	[1,11]	$[10^{(-7)}, 10^{(-6)}]$

Notes: Ranges for newly introduced B-cell parameters are either (i) derived by direct experimental fold-changes (ADCC), (ii) based on reported physiological ranges of B-cell counts, or (iii) chosen by analogy to corresponding CIR parameters in the baseline model for sensitivity analyses.

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